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# Selective Adversarial Causal Oversight: An Information-Partitioned Benchmark Contract and Countermodel-Grounded Reference Verifier

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## Abstract

LLMs are increasingly used to evaluate causal claims, but real causal oversight is both *selective* and *adversarial*: claimants may hold private information, exploit non-identifiability, and present persuasive but incomplete arguments. We formalize this setting as *selective adversarial causal oversight*, where a verifier with only public evidence must decide whether a natural-language claim is VALID, INVALID, or UNIDENTIFIABLE. We present a benchmark contract and prototype synthetic core with schema-level public/gold partitioning, twelve graph families across Pearl’s hierarchy, structurally grounded attacks, a persuasion overlay, and OOD buckets for graph, mechanism, attack-family, context-shift, and paired-flip shifts. We also propose *countermodel-grounded selective verification*, a reference pipeline that parses the claim, builds an assumption ledger, searches for observationally compatible countermodels, and returns an abstention-aware verdict. On a fixed-code exploratory snapshot (three seeds, ten samples per family, difficulty 0.55), the reference verifier improves the accuracy/unsafe-acceptance/wise-refusal tradeoff over judge-style baselines, while a deliberately leaking positive control shows how oracle information can inflate benchmark scores. These results support the need for public/gold partitioning in causal oversight benchmarks; larger synthetic runs, real-grounded cases, API-based model baselines, and human audit remain required before claiming a mature benchmark release.

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## 1. Introduction

LLMs are now used as judges, debate moderators, and policy explainers, but evaluating their *causal* reasoning remains awkward. Existing benchmarks largely treat causal questions as graded multiple-choice problems (Jin et al., 2023; 2024; Frohberg & Binder, 2022): the evaluator inspects an oracle structural causal model, fixes a yes/no answer, and scores the LLM’s prediction against it. This format is convenient but masks two features that dominate causal oversight in practice.

**Information asymmetry.** A claimant – a clinician, policy advocate, research summarizer, or deceptive agent – often has access to private information that the verifier does not: the data-generating process, hidden confounders, selection mechanisms, or simply domain familiarity. The verifier sees only public evidence and the claim text. Ignoring this asymmetry collapses a hard *oversight* problem into an easy *closed-world recall* problem.

**Wise refusal.** A core lesson of causal inference is that, given a fixed public evidence base, many queries are not identifiable (Pearl, 2009; Bareinboim et al., 2022). In such cases the correct behaviour is not to commit to a verdict, but to abstain with a reason. Modern LLM judges, however, are trained to commit and exhibit sycophancy and position bias when uncertain (Sharma et al., 2024; Chen et al., 2024; Ye et al., 2024). Existing causal benchmarks rarely measure “did the model refuse for the right reason” as a first-class quantity.

**Our task.** We formalize *selective adversarial causal oversight* (Section 3). An attacker generates a natural-language claim with full knowledge of the gold structural causal model (SCM); a verifier sees only the public projection of the scenario plus the claim text; and the evaluator scores the verifier’s *three-way* verdict {VALID, INVALID, UNIDENTIFIABLE} together with a structured assumption ledger, refusal reason, and witness.

**Our benchmark contract.** We construct a prototype synthetic core of twelve graph families that explicitly

mark family-level identifiability, observationally equivalent counter-structures, and a finite catalogue of structural attack modes. A persuasion overlay projects each claim into authority, expert-tone, confidence, consensus and concealment styles. Splits enforce partitioned holdouts at the family, mechanism, attack-family, context-shift, and paired-flip levels, and leakage-prone metadata fields are stripped from the verifier-side view by construction (Section 4). Following our current research plan, the literature-grounded subset and dual human audit are treated as required completion gates rather than finished evidence.

**Our reference verifier.** *Countermodel-grounded selective verification* (Section 5) is a four-stage pipeline that (i) parses the claim into structured slots, (ii) builds an assumption ledger of identifying preconditions, (iii) searches for observationally compatible counter-SCMs that disagree with the claim on the target query, and (iv) consults Pearl-style identification tools as supporting evidence. A fixed decision rule turns these artifacts into one of the three verdicts plus an abstention reason and a missing-information specification.

**Our claims.** Three propositions (Section 6) tie selective abstention, countermodel witnesses, and oracle leakage to formal properties of the task. Empirically (Section 7), on the current three-seed, ten-samples-per-family workspace snapshot we observe that (a) the full system improves verdict accuracy over DIRECT JUDGE from 0.504 to 1.000 on `test_iid` and from 0.161 to 0.891 on `test_ood`, while driving unsafe acceptance from 0.89/0.97 to 0.00/0.00; (b) a deliberate oracle-leaking positive control reaches 1.00 accuracy on both dedicated attack-only splits and therefore remains a useful positive control for any benchmark that does not enforce a public/gold partition; and (c) removing *abstention* preserves high IID raw accuracy (0.856) but collapses unidentifiable-awareness to 0.00 on both splits, which is exactly the failure mode the task is designed to surface. We discuss limitations – in particular, the still-modest benchmark scale, missing real-grounded human audit, and API model-baseline status – in Section 7 and Section 8.

## 2. Related Work

**Causal benchmarks for LLMs.** CLADDER (Jin et al., 2023) introduces a 10K-sample symbolic causal benchmark spanning Pearl’s three rungs, with ground truth derived from an oracle inference engine. CORR2CAUSE (Jin et al., 2024) probes whether LLMs can convert correlational statements into causal structure; the authors report near-random performance and a sharp drop under perturbation. CRASS (Frohberg & Binder, 2022) targets counterfactual conditionals, and CAUSALREASONINGBENCHMARK (Sawarni et al., 2026) separates causal identification from numerical estima-

tion on real datasets. We differ from these benchmarks in three structural ways: (i) we strip oracle fields from the verifier’s view by schema construction rather than by prompt hygiene, (ii) we elevate UNIDENTIFIABLE to a first-class supervised label tied to public evidence, and (iii) we evaluate not just accuracy but a wise-refusal-aware metric bundle and a deliberately leaking positive control.

**Fact verification and real-grounding sources.** CHECK-WHY (Si et al., 2024) studies causal fact verification with claim-evidence-argument triplets and support/refute/not-enough-info labels, while SCIFACT (Wadden et al., 2020) and HEALTHVER (Sarrouiti et al., 2021) focus on scientific and health claim verification. Clinical evidence corpora such as Evidence Inference 2.0 (DeYoung et al., 2020) and EVIDENCEOUTCOMES (Zhou et al., 2025) provide treatment-comparator-outcome language, and causal/uplift datasets such as ACIC challenges (Atlantic Causal Inference Conference, 2019), REALCAUSE (Neal et al., 2021), Criteo Uplift (Criteo AI Lab, 2014), KUAIREC/KUAI RAND (Gao et al., 2022a;b), and the Open Bandit Dataset (Saito et al., 2021) provide grounded settings with interventions, exposure policies, or logged decisions. We use these as target sources for the planned real-grounded subset, but their native labels do not directly equal our {VALID, INVALID, UNIDENTIFIABLE} causal identifiability contract and therefore require re-annotation.

**Selective prediction and abstention.** Classical selective prediction (Geifman & El-Yaniv, 2017; 2019) introduces a reject option to deep classifiers. Recent work re-derives this idea for LLMs and finds that abstention quality is sensitive to calibration (Guo et al., 2017; Wen et al., 2025). Our task definition is closest to this literature in spirit: the verifier explicitly outputs an abstention reason when public evidence does not determine the query, for example observational equivalence or missing identifying support.

**LLM-as-judge and sycophancy.** LLM judges show systematic biases: sycophancy (Sharma et al., 2024), position and verbosity bias (Ye et al., 2024), and authority effects (Chen et al., 2024). Causal claims compound these biases because the surface plausibility of an explanation is correlated with the structural correctness of the underlying identification. Our persuasion overlay is designed to expose exactly that gap.

**Scalable oversight and debate.** Irving et al. (2018) and Bowman et al. (2022) frame oversight as a game between informed agents and a less-informed judge. Our setting is a specialization of that frame to causal claims, with a structured rather than free-form transcript and an explicit abstention option for the judge.

**Causal identification and tools.** We build on canonical identification machinery: do-calculus and the backdoor/frontdoor criteria (Pearl, 2009; Shpitser & Pearl, 2008), LATE-style instrumental-variable identification (Imbens & Angrist, 1994), and probability-of-causation results (Tian & Pearl, 2000). Tool execution in our pipeline uses estimators in the spirit of DoWhy (Sharma & Kiciman, 2020).

### 3. Task: Selective Adversarial Causal Oversight

#### 3.1. Information Partition

A scenario is generated as a structural causal model  $\mathcal{M} = (\mathbf{V}, \mathbf{U}, F, P(\mathbf{U}))$  over observed variables  $\mathbf{V}$  and hidden variables  $\mathbf{U}$ . We expose three contracted views:

- *Attacker view:* the gold SCM, the true DAG, hidden variable identities, full data, and a generator metadata bundle. The attacker uses this to craft a claim that is rhetorically plausible against public evidence but structurally constrained.
- *Verifier view:* a `PublicCausalInstance` containing only observed variables, public observed data, optional named role hints (proxy/instrument/mediator/selection), and the natural-language claim.
- *Evaluator view:* gold + verifier views, used solely to score the verdict and audit the witnesses.

The contract is enforced at the schema level: the `PublicCausalInstance` dataclass strips gold-only fields such as the true DAG, hidden variables, full SCM, full data, and gold labels, and sanitizes public text against explicit leakage patterns before the verifier-side view is emitted.

#### 3.2. Output and Label Spaces

For a parsed claim with target  $(X, Y)$  over query type  $q \in \{\text{association, intervention, counterfactual}\}$ , the verifier outputs

- a verdict  $\hat{y} \in \{\text{VALID, INVALID, UNIDENTIFIABLE}\}$ ,
- an identification status  $s$  that is either IDENTIFIED, CONTRADICTED, or UNDERDETERMINED, consistent with  $\hat{y}$ ,
- a refusal reason  $r$  when  $\hat{y} = \text{UNIDENTIFIABLE}$ ,
- an assumption ledger of typed entries (SUPPORTED, CONTRADICTED, UNRESOLVED),

- witness slots for supporting evidence, countermodels, and assumptions, and
- a missing-information spec listing the unsupported assumptions and the evidence that would close the identification gap.

#### 3.3. Why the Third Label is Necessary

For three of our twelve graph families (`l1_latent_confounding`, `l2_invalid_iv`, `l3_counterfactual_ambiguity`), there exist multiple SCMs  $\mathcal{M}_1, \mathcal{M}_2$  that are observationally equivalent but disagree on the target query. Pearl’s causal hierarchy theorem (Bareinboim et al., 2022) formalizes the general case: layer- $k$  data underdetermines layer- $(k + 1)$  truth almost everywhere. We therefore treat UNIDENTIFIABLE not as a degraded label but as the unique correct output when public evidence does not pin down a single query value.

### 4. Benchmark

#### 4.1. Synthetic Core: Twelve Graph Families

The benchmark generator instantiates each scenario from a registry of twelve deterministic graph family blueprints (Table 1), distributed across the three rungs of Pearl’s hierarchy (four L1, five L2, three L3 families). Each family is tagged with a family-level identifiability status (IDENTIFIABLE vs. POTENTIALLY\_UNIDENTIFIABLE), a list of supported gold labels, and a catalogue of attack modes that match the family’s structure.

#### 4.2. Persuasion Overlay

A separate overlay layer projects each structurally-fixed claim into a persuasion-styled variant. The taxonomy mirrors the social-science persuasion literature (Zeng et al., 2024) but is deliberately narrowed to five axes: AUTHORITY, EXPERT\_TONE, CONFIDENCE, CONSENSUS, and CONCEALMENT. The paper-facing primary axis uses four of these and reports EXPERT\_TONE only in the taxonomy-complete artifact, to avoid double-counting tone effects with authority effects in the main table.

#### 4.3. Witnesses

Every gold sample carries up to three structured witnesses: a support witness (why the gold label holds under public evidence), a countermodel witness (an observationally compatible alternative SCM that disagrees with the claim on the target query), and an assumption witness (which identifying assumption is unsupported or contradicted). These witnesses are first-class evaluation objects rather than narrative decoration: they ground the decision rule in Section 5

Table 1. Twelve graph families across Pearl’s hierarchy. Each family fixes observable roles (treatment, outcome, context, adjuster, instrument, mediator, proxy, selection, latent confounder), a supported label set, and an attack catalogue. “ID” = structurally identifiable; “PU” = potentially unidentifiable.

| RUNG | FAMILY                   | ID/PU | SUPPORTED LABELS        | PRIMARY ATTACK MODES                       |
|------|--------------------------|-------|-------------------------|--------------------------------------------|
| L1   | LATENT_CONFOUNDING       | PU    | INVALID, UNIDENTIFIABLE | ASSOC. OVERCLAIM, HIDDEN-CONF. DENIAL      |
| L1   | SELECTION_BIAS           | PU    | INVALID, UNIDENTIFIABLE | SELECTION-BIAS OBFUSCATION                 |
| L1   | PROXY_DISAMBIGUATION     | ID    | VALID, INVALID          | ASSOC. OVERCLAIM                           |
| L1   | REVERSE_CAUSALITY        | PU    | INVALID, UNIDENTIFIABLE | ASSOC. OVERCLAIM                           |
| L2   | VALID_BACKDOOR           | ID    | VALID, INVALID          | INVALID-ADJUSTMENT, HETEROGENEITY OVERGEN. |
| L2   | VALID_IV                 | ID    | VALID, INVALID          | HETEROGENEITY OVERGEN.                     |
| L2   | INVALID_IV               | PU    | INVALID, UNIDENTIFIABLE | WEAK-IV-AS-VALID-IV, EXCLUSION CLAIM       |
| L2   | SUBGROUP_HETEROGENEITY   | PU    | INVALID, UNIDENTIFIABLE | HETEROGENEITY OVERGEN.                     |
| L2   | FRONTDOOR_PARTIAL_MEAS.  | PU    | VALID, UNIDENTIFIABLE   | UNIDENTIFIABLE DISGUISED AS VALID          |
| L3   | COUNTERFACTUAL_AMBIGUITY | PU    | UNIDENTIFIABLE          | COUNTERFACTUAL OVERCLAIM                   |
| L3   | MEDIATION_ABDUCTION      | ID    | VALID, INVALID          | FUNCTION-FORM MANIPULATION                 |
| L3   | MONOTONICITY_CROSS_WORLD | PU    | INVALID, UNIDENTIFIABLE | CROSS-WORLD FAILURE, FUNCTION-FORM         |

and the qualitative witness-faithfulness probe in Section 7.

#### 4.4. Splits and Holdouts

The split manifest enforces non-overlapping `train`, `dev`, `test_iid`, and `test_ood` subsets. The holdout strategy explicitly tracks several orthogonal axes: graph family, mechanism (e.g. confounding vs. collider), attack-family, context-shift domain, paired-flip pairs, plus lexical/template and variable-renaming controls inside the generator. This goes beyond the lexical-perturbation OOD studied in prior work (Jin et al., 2024), which we view as a useful but insufficient stress test. Section 7 reports the five paper-facing OOD buckets that correspond directly to the benchmark contract: graph-family, mechanism, attack-family, context-shift, and paired-flip OOD.

#### 4.5. Real-Grounded Subset

A literature-grounded subset (target size 150–300 cases) is curated from economics, policy, epidemiology and education observational studies. Each case records a source citation, a public-evidence summary, the visible-vs-hidden information contract, the gold label, identifying assumptions, and a witness note. The synthetic and real-grounded subsets share the same claim schema, so models trained or tuned on the synthetic core can be zero-shot evaluated on the real-grounded subset. This subset is part of the benchmark *design*; the current workspace also includes a 150-case human-audit package generator, but not yet dual-annotator agreement results. Full real-grounded evaluation is therefore reserved for the human-audit phase (see Section 8).

### 5. Method: Countermodel-Grounded Selective Verification

The verifier pipeline runs four sequential stages and emits a single `VerifierDecision` object.

**Stage 1: Claim parsing.** Given a public scenario and the natural-language claim, the parser extracts treatment, outcome, query type, polarity, strength, mentioned and implied assumptions, the rhetorical strategy (e.g. an adjustment-sufficiency claim or an IV appeal), and an abstention-risk flag derived from claim-level cues.

**Stage 2: Assumption ledger.** The ledger initializes one entry per identifying precondition required by the parsed claim. Status starts as `UNRESOLVED` and is updated by later stages: tool-backed evidence may flip an entry to `SUPPORTED`, contradictory evidence flips it to `CONTRADICTED`. Baseline assumptions (`consistency`, `positivity`) are tracked but not counted as identifying support.

**Stage 3: Countermodel search.** The countermodel module enumerates structurally-compatible alternative SCMs for the public scenario and scores each on (a) observational match score, (b) disagreement with the claim on the target query, (c) which identifying assumption it triggers. The module returns a chosen candidate, a short explanation, and a verdict suggestion.

**Stage 4: Tool-backed adjudication.** When no countermodel survives, a tool runner executes the appropriate identification tools for the rung – conditional independence and association checks for L1, backdoor, frontdoor and IV estimators for L2, and SCM-based counterfactual reasoning plus probability-of-necessity bounds for L3 (Tian & Pearl, 2000; Sharma & Kiciman, 2020). Tool outputs are stored as a structured trace that records which ledger entries are supported or contradicted.

**Decision rule.** The four stages are combined by a fixed rule:

1. If a strong countermodel survives and the claim’s verdict suggestion is `INVALID`, return `INVALID`.



2. Else if a countermodel survives and the candidates disagree on the target query, return UNIDENTIFIABLE with refusal reason `observational-equivalence`.
3. Else if any identifying assumption is contradicted by tools, return INVALID.
4. Else if any core identifying assumption remains unsupported or no primary-supporting tool record exists, return UNIDENTIFIABLE with a missing-identifying-support reason; when the abstention gate fires on a high-risk claim, we return the stricter primary-support variant.
5. Else return VALID.

This rule is deterministic, auditable, and is implemented in fewer than 300 lines of Python; the closed-form schema makes selective verdicts directly comparable across systems.

## 6. Theoretical Guarantees

We make three minimal claims that pin down why the task design is necessary rather than aesthetic.

**Proposition 6.1** (Abstention Necessity). *Let  $E$  denote the public evidence and  $q$  the target query. Suppose there exist SCMs  $\mathcal{M}_1, \mathcal{M}_2$  that both induce  $E$  as their observational marginal but disagree on  $q$ . Then any verifier that does not admit UNIDENTIFIABLE as an output produces an incorrect verdict on at least one of  $\mathcal{M}_1, \mathcal{M}_2$ .*

*Proof sketch.* Both SCMs are public-evidence-equivalent, so the verifier’s input distribution is identical under either generative model. Any committed verdict is therefore the same under both, but  $q(\mathcal{M}_1) \neq q(\mathcal{M}_2)$ , so the verdict is wrong under exactly one.  $\square$

*Empirical correspondence.* Section 7 (NO-ABSTENTION ablation) instantiates this proposition: removing the abstention head retains 0.856/0.462 raw accuracy on IID/OOD but zeroes out unidentifiable-awareness.

**Proposition 6.2** (Countermodel Witness Soundness). *Suppose the verifier exhibits an SCM  $\mathcal{M}'$  that (a) is consistent with  $E$  up to an observational match score  $s \geq 1 - \varepsilon$  and (b) disagrees with the claim on  $q$ . Then under standard SCM identifiability (Shpitser & Pearl, 2008), the claim cannot be uniquely determined by  $E$  alone for any  $\varepsilon$  such that  $\mathcal{M}'$  remains in the equivalence class.*

*Proof sketch.* If  $\mathcal{M}'$  matches  $E$  to within the equivalence-class tolerance and disagrees with the claim on  $q$ , then  $E$  is consistent with at least two distinct values of  $q$ , contradicting any claim that  $E$  alone uniquely determines  $q$ . The qualitative shuffle/delete probe in Section 7 provides exploratory empirical support.  $\square$

**Proposition 6.3** (Oracle Leakage Inflation). *Let  $V_{\text{public}}$  and  $V_{\text{leak}}$  be two verifiers that differ only in that  $V_{\text{leak}}$  has read access to a gold-only field  $g \in \text{GOLD\_ONLY\_FIELDS}$ . Then  $\mathbb{E}[\text{Acc}(V_{\text{leak}})] \geq \mathbb{E}[\text{Acc}(V_{\text{public}})]$  for an oracle-positive-control verifier that can decode label signal from  $g$ , with strict inequality whenever the decoded signal changes at least one expected error of  $V_{\text{public}}$  into a correct prediction.*

*Empirical correspondence.* Section 7 (Leakage Study) instantiates this proposition with a positive-control verifier; the artifact’s `supports_warning` field operationalizes the proposition as a benchmark-level contamination gate.

The role of Theorem 6.3 is operational: it justifies our leakage study (Section 7) by predicting that any inflation observed there is, by construction, leakage rather than oversight ability.

## 7. Experiments

### 7.1. Experimental Setup

We report the current workspace snapshot: three seeds with ten samples per family across all twelve graph families. The main benchmark runs at difficulty 0.55 and compares the full *countermodel-grounded* verifier against judge-style, debate-style, tool-only and refusal-aware baselines. The component ablation suite replaces one or more pipeline stages with a no-op surrogate. We also include a dedicated attack-only leakage study under the hidden-information-aware profile. These runs are reproducible from the current repository state, but they are not a saturation study: the synthetic core is still modest, the leakage study is exploratory because it uses ten rather than the pre-registered sixty samples per family, and the real-grounded, human-audit and API-based model-baseline phases are not yet complete. We therefore present the empirical results as fixed-code evidence for the benchmark contract and reference verifier, not as a final benchmark release.

### 7.2. Main Benchmark

Table 2 compares the full COUNTERMODEL-GROUNDED verifier against compact judge-style and system-style baselines on `test_iid` and `test_ood`. We report the three metrics most central to our task definition: verdict accuracy, unsafe-acceptance rate, and wise-refusal recall. SELF-CONSISTENCY JUDGE matches COT JUDGE exactly in this snapshot and is omitted from the compact table for readability; the full artifact retains it.

Three observations matter. First, DIRECT JUDGE exhibits exactly the failure mode that motivates the task: extremely high unsafe acceptance (0.89 IID, 0.97 OOD) under public evidence constraints. Second, COT JUDGE, DEBATE BASELINE, and the explicit REFUSAL-AWARE baseline drive

Table 2. Main benchmark, current mainline snapshot (three seeds, ten samples per family, difficulty 0.55). Mean  $\pm$  std across seeds; full 95% CIs, calibration metrics, and over-refusal rates are reported in `outputs/review.fixed.exp.main.benchmark.md`.

| SYSTEM                | SPLIT    | VERDICT ACC.      | UNSAFE ACCEPT     | WISE REFUSAL RECALL |
|-----------------------|----------|-------------------|-------------------|---------------------|
| DIRECT JUDGE          | TEST_IID | 0.504 $\pm$ 0.090 | 0.889 $\pm$ 0.192 | 0.500 $\pm$ 0.500   |
| DIRECT JUDGE          | TEST_OOD | 0.161 $\pm$ 0.056 | 0.970 $\pm$ 0.029 | 0.184 $\pm$ 0.034   |
| CoT JUDGE             | TEST_IID | 0.322 $\pm$ 0.019 | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000   |
| CoT JUDGE             | TEST_OOD | 0.592 $\pm$ 0.053 | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000   |
| TOOL BASELINE         | TEST_IID | 0.54 $\pm$ 0.12   | 0.37 $\pm$ 0.26   | 0.67 $\pm$ 0.58     |
| TOOL BASELINE         | TEST_OOD | 0.346 $\pm$ 0.092 | 0.423 $\pm$ 0.057 | 0.560 $\pm$ 0.067   |
| DEBATE BASELINE       | TEST_IID | 0.322 $\pm$ 0.019 | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000   |
| DEBATE BASELINE       | TEST_OOD | 0.456 $\pm$ 0.040 | 0.000 $\pm$ 0.000 | 0.610 $\pm$ 0.026   |
| REFUSAL-AWARE         | TEST_IID | 0.752 $\pm$ 0.045 | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000   |
| REFUSAL-AWARE         | TEST_OOD | 0.504 $\pm$ 0.090 | 0.000 $\pm$ 0.000 | 0.591 $\pm$ 0.020   |
| COUNTERMODEL-GROUNDED | TEST_IID | 1.000 $\pm$ 0.000 | 0.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000   |
| COUNTERMODEL-GROUNDED | TEST_OOD | 0.891 $\pm$ 0.005 | 0.000 $\pm$ 0.000 | 0.825 $\pm$ 0.018   |

unsafe acceptance to zero, but do so by over-refusing: on `test_ood`, their over-refusal rates are 0.850, 0.677, and 0.557, respectively, versus 0.010 for the full system. Third, the near-perfect IID score should be read as a reference-verifier sanity check on the synthetic contract, not as evidence that generic LLMs have solved causal oversight. The stronger claim is the tradeoff shape: the full pipeline maintains zero unsafe acceptance while preserving OOD wise-refusal recall and substantially improving raw verdict accuracy over public judge-style baselines.

### 7.3. Component Ablation: What Drives Wise Refusal?

Table 3 extends the ablation to four pipeline components, including a NO-ABSTENTION variant that disables the third-label output entirely. Two patterns are especially important. First, NO-ABSTENTION preserves high IID raw accuracy (0.856) but zeroes out unidentifiable-awareness on both splits, which is exactly the failure mode the selective task is designed to surface. Second, removing the countermodel stage hurts both OOD verdict accuracy and OOD unidentifiable-awareness (0.504 and 0.591). The current snapshot is not a strict monotone ranking across every metric: NO-TOOLS exceeds the full system on OOD raw accuracy (0.911 vs. 0.891) and unidentifiable-awareness (0.964 vs. 0.825), but does so with substantially higher over-refusal (0.150 vs. 0.010) and lower wise-refusal precision (0.877 vs. 0.989). We therefore interpret the full system as the more balanced reference verifier, not as uniformly best on every scalar metric.

### 7.4. Leakage Study

The current leakage study is explicitly exploratory because it uses only ten samples per family. We therefore present it as a positive-control sanity check that the public/gold contract behaves as Proposition 6.3 predicts, rather than as a definitive estimate of contamination magnitude. Full

Table 3. Component ablation. Unidentifiable-awareness is the task-critical third-label measure; the full artifact includes precision, over-refusal, calibration, and countermodel coverage.

| SYSTEM          | SPLIT | ACC.  | UNIDENT. AWARENESS |
|-----------------|-------|-------|--------------------|
| FULL            | IID   | 1.000 | 1.000              |
| FULL            | OOD   | 0.891 | 0.825              |
| NO-LEDGER       | IID   | 0.144 | 1.000              |
| NO-LEDGER       | OOD   | 0.520 | 1.000              |
| NO-COUNTERMODEL | IID   | 0.752 | 1.000              |
| NO-COUNTERMODEL | OOD   | 0.504 | 0.591              |
| NO-ABSTENTION   | IID   | 0.856 | 0.000              |
| NO-ABSTENTION   | OOD   | 0.462 | 0.000              |
| NO-TOOLS        | IID   | 0.570 | 1.000              |
| NO-TOOLS        | OOD   | 0.911 | 0.964              |

leakage evidence requires increasing `samples_per_family` to at least 60.

We instantiate two verifiers on a dedicated attack-only benchmark with the hidden-information-aware attack profile: a clean public verifier and a deliberately oracle-leaking control that decodes benchmark-answer supervision from a protocol-violating public metadata partition. Per Theorem 6.3, any positive gap is *by construction* leakage inflation. Table 4 shows a gap of +0.421 on `test_iid` and +0.155 on `test_ood`. Both paired tests support the warning in this small run (bootstrap  $p = 0.0395/0.0015$ , McNemar  $p = 0.0156/0.0000$  before the table’s rounding), but the protocol notice still prevents treating this as formal leakage evidence until the sample count is increased. ECE drops from 0.400 to 0.000 on `test_iid` and from 0.077 to 0.000 on `test_ood`, illustrating how leaked supervision can masquerade as calibration.

### 7.5. Adversarial Robustness

Table 5 reports robustness across four attack-strength regimes: WEAK, MEDIUM, STRONG, and a leak-aware hidden-information profile. On `test_ood`, the verifier

Table 4. Leakage study. Clean public verifier vs. an oracle-positive control with protocol-violating public metadata. Any positive gap is leakage inflation (Theorem 6.3). Numbers from the accompanying outputs/review\_fixed\_exp\_leakage\_study.md artifact.

| METRIC (TEST_IID)      | CLEAN | LEAKING | $\Delta$ |
|------------------------|-------|---------|----------|
| ACCURACY               | 0.579 | 1.000   | +0.421   |
| MACRO-F1               | 0.373 | 0.667   | +0.294   |
| ECE ( $\downarrow$ )   | 0.400 | 0.000   | -0.400   |
| BRIER ( $\downarrow$ ) | 0.220 | 0.000   | -0.220   |
| METRIC (TEST_OOD)      | CLEAN | LEAKING | $\Delta$ |
| ACCURACY               | 0.845 | 1.000   | +0.155   |
| MACRO-F1               | 0.563 | 0.667   | +0.104   |
| ECE ( $\downarrow$ )   | 0.077 | 0.000   | -0.077   |
| BRIER ( $\downarrow$ ) | 0.086 | 0.000   | -0.086   |

Table 5. Adversarial robustness. Mean  $\pm$  std across seeds; full results are reported in the accompanying exp\_adversarial\_robustness.md artifact.

| STRENGTH   | SPLIT | ACC.              | WISE REFUSAL RECALL |
|------------|-------|-------------------|---------------------|
| WEAK       | IID   | 0.889 $\pm$ 0.192 | 0.667 $\pm$ 0.577   |
| WEAK       | OOD   | 0.845 $\pm$ 0.039 | 0.685 $\pm$ 0.082   |
| MEDIUM     | IID   | 0.889 $\pm$ 0.192 | 0.667 $\pm$ 0.577   |
| MEDIUM     | OOD   | 0.845 $\pm$ 0.039 | 0.685 $\pm$ 0.082   |
| STRONG     | IID   | 0.889 $\pm$ 0.192 | 0.667 $\pm$ 0.577   |
| STRONG     | OOD   | 0.845 $\pm$ 0.039 | 0.685 $\pm$ 0.082   |
| LEAK-AWARE | IID   | 0.579 $\pm$ 0.417 | 0.667 $\pm$ 0.577   |
| LEAK-AWARE | OOD   | 0.845 $\pm$ 0.067 | 0.844 $\pm$ 0.104   |

stays in a narrow 0.845 band across all four regimes while maintaining zero invalid acceptance. The `test_iid` picture is less stable: the hidden-information-aware regime drops to  $0.579 \pm 0.417$ , making it the hardest current IID stress case. Because the weak/medium/strong rows are nearly identical in this snapshot, we treat this section as exploratory stress testing rather than a headline robustness claim.

## 7.6. OOD Buckets

The current OOD artifact evaluates the five buckets defined by the benchmark contract: graph-family, mechanism, attack-family, context-shift, and paired-flip OOD. The full system remains perfect on graph-family and mechanism OOD (1.000 accuracy), stays strong on attack-family (0.820) and context-shift (0.883), and drops sharply on paired-flip (0.439). The paired-flip bucket also has the largest macro-F1 gap (0.423), making it the clearest current stress case for structural disambiguation. The current significance artifact reports uninformative  $p = 1$  rows for these bucket gaps, so we avoid strong significance claims until the OOD estimand and sample count are upgraded.

## 7.7. Witness Faithfulness (qualitative)

We run a small witness-faithfulness probe by shuffling or deleting the countermodel witness in the verifier output and re-scoring. When the countermodel witness is shuffled across instances, the system’s unidentifiable-awareness collapses on the families that rely on observational-equivalence reasoning (`l1_latent_confounding_family`, `l3_counterfactual_ambiguity_family`). This indicates that the witness is load-bearing rather than ornamental. A full quantitative study (witness shuffle/delete with paired-bootstrap 95% CIs across all twelve families) is reserved for future work.

## 8. Discussion and Limitations

**What we are not claiming.** We are *not* claiming a state-of-the-art result on a saturated benchmark, nor that LLM judges are uniformly bad at causal reasoning. The claim is structural: once the task enforces a public/gold contract and rewards correct abstention, the ranking of systems changes, and a verifier that takes this shape seriously offers a better accuracy/refusal tradeoff than verifiers that flatten the problem into a closed-world judgment task.

**Limitations.** The synthetic core is, by construction, a model of the world rather than the world itself. The real-grounded subset is small, and full human dual-annotation is in progress. The current mainline snapshot uses ten samples per family, so bucket-level estimates can still have wide CIs; the dedicated leakage study is explicitly exploratory; and the hidden-information-aware IID robustness slice is not yet stable enough to support strong quantitative claims. Larger runs and completed human audit will be needed before any broad statement about absolute performance is defensible. The existing transfer matrix is likewise a pilot: it shows that family-swapped attacker/verifier profiles are testable inside the benchmark contract, but it is still too small to support model-family conclusions. We document these caveats in the artifact and in this paper.

**Future work.** Two priorities: (i) closing the real-grounded subset with dual annotation and Cohen’s  $\kappa$  on the four primary audit fields; (ii) scaling the current transfer pilot into a full-budget cross-model study across verifier families and sizes, which will let us connect the structural conclusions of this paper to specific model regimes.

## 9. Conclusion

Selective adversarial causal oversight is the right unit of analysis for LLM-based causal evaluation. By modelling information asymmetry, treating UNIDENTIFIABLE as a first-class label, partitioning public from gold at the schema level,

and grounding the verifier in countermodel witnesses, we obtain a benchmark and a method whose comparison shape is meaningful even under current mainline budgets. The next step is scale: more samples, more graph families, real-grounded human-audited cases, and cross-model transfer.

## Software and Data

The repository contains the benchmark generator, verifier pipeline, experiment runners, fixed-code result artifacts, and regression tests used for the current snapshot. A paper-facilitating reproducibility package will be released after the real-grounded subset, human audit, and API-model baseline runs are completed. Each experiment emits a configuration JSON, a seed list, raw per-sample predictions, an aggregated metrics file, 95% CIs, paired-significance tests, and a markdown summary. The leakage study additionally emits a leakage-warning field that downstream consumers can use as a contamination check.

## Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here. We note in particular that our adversarial framing is intended for evaluation and oversight rather than deployment: the persuasion overlay is a stress-test artifact for verifiers, not a recipe for deceptive content generation. The leakage decoder is isolated as a positive-control diagnostic so that contamination risks are measurable rather than hidden.

## References

Atlantic Causal Inference Conference. ACIC 2019 data challenge, 2019. URL <https://sites.google.com/view/ACIC2019DataChallenge/data-challenge>.

Bareinboim, E., Correa, J. D., Ibeling, D., and Icard, T. On Pearl’s hierarchy and the foundations of causal inference. In Geffner, H., Dechter, R., and Halpern, J. Y. (eds.), *Probabilistic and Causal Inference: The Works of Judea Pearl*, pp. 507–556. ACM Books, 2022. doi: 10.1145/3501714.3501743.

Bowman, S. R., Hyun, J., Perez, E., Chen, E., Pettit, C., Heiner, S., Lukosiute, K., Askell, A., Jones, A., Chen, A., et al. Measuring progress on scalable oversight for large language models. *arXiv preprint arXiv:2211.03540*, 2022.

Chen, G. H., Chen, S., Liu, Z., Jiang, F., and Wang, B. Humans or LLMs as the judge? a study on judgement

bias. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.

Criteo AI Lab. Criteo uplift prediction dataset, 2014. URL <https://ailab.criteo.com/criteo-uplift-prediction-dataset/>.

DeYoung, J., Lehman, E., Nye, B., Marshall, I. J., and Wallace, B. C. Evidence inference 2.0: More data, better models. In *Proceedings of the 19th SIGBioMed Workshop on Biomedical Language Processing*, 2020. URL <https://aclanthology.org/2020.bionlp-1.13/>.

Frohberg, J. and Binder, F. CRASS: A novel data set and benchmark to test counterfactual reasoning of large language models. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference (LREC)*, pp. 2126–2140, 2022.

Gao, C., Li, S., Lei, W., Chen, J., Li, B., Jiang, P., He, X., Mao, J., and Chua, T.-S. KuaiRec: A fully-observed dataset and insights for evaluating recommender systems. In *Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM)*, 2022a.

Gao, C., Li, S., Zhang, Y., Chen, J., Li, B., Lei, W., Jiang, P., and He, X. KuaiRand: An unbiased sequential recommendation dataset with randomly exposed videos. In *Proceedings of the 31st ACM International Conference on Information and Knowledge Management (CIKM)*, pp. 3953–3957, 2022b. doi: 10.1145/3511808.3557624. URL <https://kuairand.com/>.

Geifman, Y. and El-Yaniv, R. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Geifman, Y. and El-Yaniv, R. SelectiveNet: A deep neural network with an integrated reject option. In *International Conference on Machine Learning (ICML)*, 2019.

Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, 2017.

Imbens, G. W. and Angrist, J. D. Identification and estimation of local average treatment effects. *Econometrica*, 62 (2):467–475, 1994.

Irving, G., Christiano, P., and Amodei, D. AI safety via debate. *arXiv preprint arXiv:1805.00899*, 2018.

Jin, Z., Chen, Y., Leeb, F., Gresele, L., Kamal, O., Lyu, Z., Blin, K., Adauro, F. G., Kleiman-Weiner, M., Sachan, M., and Schölkopf, B. CLadder: Assessing causal reasoning in language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.



- Jin, Z., Liu, J., LYU, Z., Poff, S., Sachan, M., Mihalcea, R., Diab, M. T., and Schölkopf, B. Can large language models infer causation from correlation? In *International Conference on Learning Representations (ICLR)*, 2024.
- Lanham, T., Chen, A., Radhakrishnan, A., Steiner, B., Denison, C., Hernandez, D., Li, D., Durmus, E., Hubinger, E., Kernion, J., Lukosiute, K., Nguyen, K., Cheng, N., Joseph, N., Schiefer, N., Rausch, O., Larson, R., McCandlish, S., Kundu, S., Kadavath, S., Yang, S., Henighan, T., Maxwell, T., Telleen-Lawton, T., Hume, T., Hatfield-Dodds, Z., Kaplan, J., Brauner, J., Bowman, S. R., and Perez, E. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023.
- Neal, B., Huang, C.-W., and Raghupathi, S. RealCause: Realistic causal inference benchmarking. *arXiv preprint arXiv:2011.15007*, 2021. URL <https://arxiv.org/abs/2011.15007>.
- Pearl, J. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.
- Pearl, J. and Mackenzie, D. *The Book of Why: The New Science of Cause and Effect*. Basic Books, 2018.
- Saito, Y., Aihara, S., Matsutani, M., and Narita, Y. Open bandit dataset and pipeline: Towards realistic and reproducible off-policy evaluation. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021. URL <https://arxiv.org/abs/2008.07146>.
- Sarrouiti, M., Abacha, A. B., Mrabet, Y., and Demner-Fushman, D. HealthVer: A dataset for evidence-based fact-checking of health claims. In *Findings of the Association for Computational Linguistics: EMNLP*, 2021. URL <https://aclanthology.org/2021.findings-emnlp.297/>.
- Sawarni, A., Tan, J., and Syrgkanis, V. CausalReasoningBenchmark: A real-world benchmark for disentangled evaluation of causal identification and estimation. *arXiv preprint arXiv:2602.20571*, 2026. URL <https://arxiv.org/abs/2602.20571>.
- Sharma, A. and Kiciman, E. DoWhy: An end-to-end library for causal inference. *arXiv preprint arXiv:2011.04216*, 2020. URL <https://arxiv.org/abs/2011.04216>.
- Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askeel, A., Bowman, S. R., Cheng, N., Durmus, E., Hatfield-Dodds, Z., Johnston, S. R., Kravec, S., Maxwell, T., McCandlish, S., Ndousse, K., Rausch, O., Schiefer, N., Yan, D., Zhang, M., and Perez, E. Towards understanding sycophancy in language models. In *International Conference on Learning Representations (ICLR)*, 2024.
- Shpitser, I. and Pearl, J. Complete identification methods for the causal hierarchy. *Journal of Machine Learning Research*, 9:1941–1979, 2008.
- Si, C., Jiang, Z., Zhu, Y., and Yang, D. CHECK-WHY: Causal fact verification via argument structure. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. URL <https://aclanthology.org/2024.acl-long.835/>.
- Tian, J. and Pearl, J. Probabilities of causation: Bounds and identification. *Annals of Mathematics and Artificial Intelligence*, 28(1–4):287–313, 2000.
- Wadden, D., Lin, S., Lo, K., Wang, L. L., van Zuylen, M., Cohan, A., and Hajishirzi, H. SciFact: A dataset for verifying scientific claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020. URL <https://arxiv.org/abs/2004.14974>.
- Wen, B., Yao, J., Feng, S., Xu, C., Tsvetkov, Y., Howe, B., and Wang, L. L. Know your limits: A survey of abstention in large language models. *Transactions of the Association for Computational Linguistics*, 13:529–556, 2025. doi: 10.1162/tacl.a.00754. URL <https://aclanthology.org/2025.tacl-1.26/>.
- Ye, J., Wang, Y., Huang, Y., Chen, D., Zhang, Q., Moniz, N., Gao, T., Geyer, W., Huang, C., Chen, P.-Y., Chawla, N. V., and Zhang, X. Justice or prejudice? Quantifying biases in LLM-as-a-judge. *arXiv preprint arXiv:2410.02736*, 2024.
- Zeng, Y., Lin, H., Zhang, J., Yang, D., Jia, R., and Shi, W. How Johnny can persuade LLMs to jailbreak them: Rethinking persuasion to challenge AI safety by humanizing LLMs. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- Zhou, Y., Newbury, A. M., Zhang, G., Idnay, B. R. S., Liu, H., Weng, C., and Peng, Y. EvidenceOutcomes: A dataset of clinical trial publications with clinically meaningful outcomes. *Studies in Health Technology and Informatics*, 329:723–727, 2025. doi: 10.3233/SHTI250935. URL <https://arxiv.org/abs/2506.05380>.

## A. Schema Excerpts

The following is an abridged view of the leakage-control contract. The full `benchmark/schema.py` file contains additional sanitizers and serialization helpers; only the contract-level constants are reproduced here.

```
GOLD_ONLY_FIELDS = (
    "true_dag", "hidden_variables", "true_scm",
    "full_data", "ground_truth", "gold_label",
)

VERIFIER_VISIBLE_FIELDS = (
    "scenario_id", "description", "variables", "proxy_variables",
    "selection_mechanism", "observed_data", "data",
    "causal_level", "sanitized_public_metadata",
)

class VerdictLabel(str, Enum):
    VALID = "valid"
    INVALID = "invalid"
    UNIDENTIFIABLE = "unidentifiable"

class IdentificationStatus(str, Enum):
    IDENTIFIED = "identified"
    CONTRADICTED = "contradicted"
    UNDERDETERMINED = "underdetermined"
```

## B. Decision Rule (Pseudocode)

```
def decide_verdict(parsed, ledger, cm, tool_trace):
    # Stage 1: strong invalid countermodel
    if cm.found and cm.suggests_invalid:
        return INVALID

    # Stage 2: observational equivalence + query disagreement
    if cm.found and cm.query_disagreement:
        return UNIDENTIFIABLE(reason="observational_equivalence")

    # Stage 3a: contradicted assumption
    if any(e.contradicted for e in ledger):
        return INVALID

    # Stage 3b: missing or unsupported identification
    if any(e.unsupported for e in ledger) \
        or not tools_support_claim(tool_trace) \
        or (needs_explicit_support and no_supported_id):
        return UNIDENTIFIABLE(reason="missing_identifying_support")

    # Stage 3c: abstention gate on high-risk valid claims
    if abstention_gate_reasons(parsed, ledger, tool_trace):
        return UNIDENTIFIABLE(reason="missing_primary_identifying_support")

    # Stage 4: support
    return VALID
```

## C. Persuasion Overlay Examples

A claim with structural label UNIDENTIFIABLE can be projected into five persuasion styles. The taxonomy-complete artifact includes EXPERT\_TONE; the paper-facing primary axis collapses to the four remaining styles to avoid double-counting tone effects.

- AUTHORITY: “As established by the foundational paper of the field, the effect of  $X$  on  $Y$  is positive.”
- CONFIDENCE: “It is unambiguous, on any reading of the data, that  $X$  causes  $Y$ .”
- CONSENSUS: “Every recent meta-analysis agrees that  $X$  raises  $Y$ .”
- CONCEALMENT: omits any mention of the latent confounder or selection mechanism that the public evidence cannot resolve.
- EXPERT TONE: heavy technical jargon without a corresponding identification argument.

## D. Reproducibility Notes

Each experiment runner emits a fixed bundle of artifacts:

```
<run>_config.json
<run>_seed_list.json
<run>_raw_predictions.jsonl
<run>_aggregated_metrics.json
<run>_ci.json
<run>.md    (paper-facing markdown summary)
```

The aggregator reports seed-mean estimates and paired bootstrap 95% CIs; the significance file applies Holm-Bonferroni correction within and across splits. The leakage runner additionally writes a `supports_warning` field that downstream consumers can use as a hard gate on contamination, in line with Theorem 6.3.